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Speckle-illuminated Fourier ptychography: analysis of diffuser roughness and reconstruction aperture on imaging performance

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With a fundamentally modified structural illumination algorithm, the recently proposed speckle-illuminated Fourier ptychography can be a promising superresolution imaging technique with a large field of view. However, its imaging performance, including image resolution and signal-to-noise ratio, has been discussed less, limiting its further applications. Thus, an in-depth study of this new imaging technique is highly required. In this paper, with theoretical analysis, numerical simulations, and experiments, the influence of both diffuser roughness in the experimental setup and numerical aperture size in iterative reconstruction on the imaging performance of speckle-illuminated Fourier ptychography was studied in detail, and the result explained why a rougher diffuser and larger reconstruction aperture can generate a higher-resolution image with more noise and showed how to get optimized diffuser roughness and reconstruction aperture size by considering the trade-off between imaging resolution and signal-to-noise ratio. This work may be a good reference for high-quality imaging using speckle-illuminated Fourier ptychography. © 2020 Optical Society of America

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1. INTRODUCTION

Optical microscopic imaging was crucially important for research in biomedicine, material science, microelectronics, etc. Spatial resolution is the most important parameter to evaluate the performance of an imaging system, and the pursuit of high resolution has always been an important goal of optical engineers since the invention of the microscope. According to Abbe theory, the resolution of a microscope is determined by its illumination wavelength and numerical aperture. Thus, there are two conventional ways to improve the imaging resolution of a microscope. One way is to use short wavelength sources as illumination, and another way is to increase the numerical aperture of the imaging system. The use of x ray and high-energy electron beams for illumination can reach a resolution at the subnanometer level, but the device is always quite complex and is not applicable for live specimens. In the regime of visible light, the highest numerical aperture reached is quite close to the theoretical maximum; unfortunately, the obtained resolution is still too low to extract precise sample details.

In past decades, many methods were proposed to improve the imaging resolution by only using objectives of moderate numerical apertures and visible light sources, such as stimulated emission depletion (STED) microscopy [1–4], photoactivation localization microscopy (PALM) [5–9], stochastic optical reconstruction microscopy (STORM) [10–13], and structured illumination microscopy (SIM) [14–18]. Among these methods, SIM, proposed by Gustafsson [14,17], employs structural light to illuminate the sample under inspection to reconstruct superresolution images with high speed using cost-effective systems. Moreover, it can be adopted in various optical microscopes, such as common bright-field microscopes [19–21], differential interference contrast microscopes [22–26] and fluorescence microscopes [27–29]. Therefore, SIM was widely used in various applications where a high-resolution image was required [15]. For SIM in the early stage, the illumination should be accurately known *a priori* to reconstruct an image with superresolution, but inevitable illumination errors in optical experiments always degraded the reconstruction accuracy. In order to solve this problem, Mudry *et al.* [16] proposed an improved method termed blind SIM to reconstruct superresolution images without knowing the structural illumination in advance. In this improved method, a diffuser was employed to

form speckle illumination on the sample under detection, and multiple speckle images were collected by shifting the sample relative to the speckle illumination; then the illumination light as well as the sample can be reconstructed iteratively, getting a solution two times higher than that of conventional wide-field microscopy. The performance of blind SIM was further improved by Guo *et al.* [30] using the Fourier ptychographic algorithm to achieve images with both high resolution and large field of view (FoV) simultaneously. Though speckle-illuminated Fourier ptychography has been successfully adopted in various applications [31–35], there is still a lack of in-depth analysis on it. For example, we find in experiments that both diffuser roughness and reconstruction aperture size can significantly influence image resolution and signal-to-noise ratio (SNR); however, there is still no analysis on how imaging quality is influenced by these factors.

In this work, we quantitatively analyzed the influence of both diffuser roughness and reconstruction aperture size on image resolution and the SNR of speckle-illuminated Fourier ptychography with theoretical analysis, numerical simulations, and experiments. We find that a rougher diffuser and larger reconstruction aperture yield higher imaging resolution and lower imaging SNR in general. In addition, optimized diffuser roughness and reconstruction aperture size can be determined by considering the trade-off between imaging resolution and SNR. It is believed that this work can be a good reference to decide diffuser roughness and reconstruction aperture to get optimized reconstruction in imaging with speckle-illuminated Fourier ptychography.

2. PRINCIPLE

The pupil function of an imaging system is equivalent to its coherent transfer function (CTF), which acts as a low-pass filter on the frequency components of the object. For a common optical imaging system, only frequency components inside its pupil can be collected by the lens, and outside high-frequency components are lost, and thus the resolution of the image generated is always lower than that of the object. Conversely, the image resolution can possibly be improved if we can move the outside high-frequency components into the optical pupil and reconstruct them with the proper algorithm, which is the basic principle of all structural illumination-based imaging techniques, including speckle-illuminated Fourier ptychography. Figure 1(A) shows the sample spectrum, and Fig. 1(B) shows the image spectrum. Only a sample spectrum within the CTF can be collected while that beyond the CTF is lost. When a diffuser is introduced in front of the sample, the spectrum of exiting light is broadened, since it is the convolution between the sample spectrum and the diffuser spectrum. Then, some high-frequency components can be collected, as shown in Fig. 1(C). With the deconvolution computation in speckle-illuminated Fourier ptychography, the image spectrum, including those high-frequency components collected by the spectral broadening effect, can be reconstructed as shown in Fig. 1(D), revealing that the effective CTF of the imaging system is broadened. Therefore, the imaging resolution can be enhanced.

According to the underlining physics illustrated in Fig. 1, it is possible to collect high-frequency components by using a

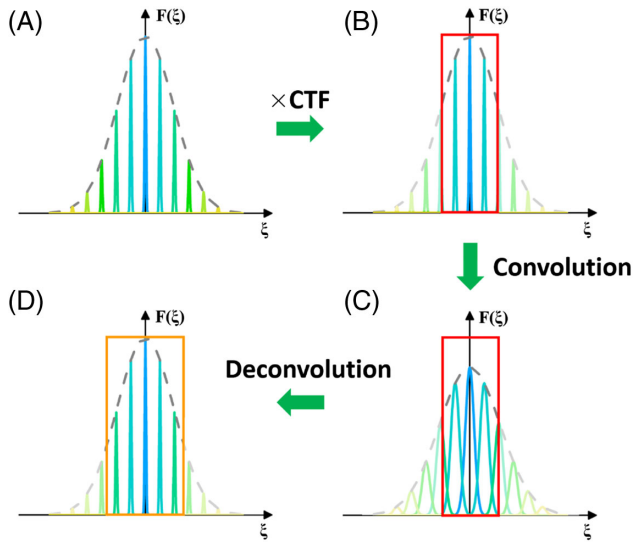

Fig. 1. Principle of speckle-illuminated Fourier ptychography. (A) Sample spectrum; (B) image spectrum and *sCTF*; (C) image spectrum with a diffuser; (D) reconstruction image spectrum and effective CTF.

diffuser, as shown in Fig. 2(A): by scanning sub-FoVs, series of raw images can be captured, and a superresolution image with a large FoV can be reconstructed via recently proposed iterative algorithms [31–35]. As discussed in various works, the diffuser roughness significantly affects imaging performance [30]; however, we found in several experiments that the diffuser roughness in the imaging system and the reconstruction aperture size in the retrieval algorithm both have significant effects on imaging performance, especially on image resolution and SNR. However, few works have considered the reconstruction aperture size in speckle-illuminated Fourier ptychography. In order to comprehensively analyze their influences, an improved reconstruction algorithm considering the effect of reconstruction aperture size is proposed, and its flow chart is illustrated in Fig. 2(B).

First, the complex amplitude $s_{m,n}(x, y)$ leaving the sample can be computed as the product of probe light $p_n(x, y)$ and sample function $o_{m,n}(x, y)$ with Eq. (1), and its spectrum $S_{m,n}(\xi, \eta)$ is computed using Fourier transform (F) in Eq. (2). The subscript m represents different sub-FoVs, and the subscript n represents the iteration number. In the first iteration $(n = 1)$, $o_{m,1}(x, y)$ can be a random initial guess:

$$s_{m,n}(x, y) = p_n(x, y) \times o_{m,n}(x, y), \tag{1}$$

$$S_{m,n}(\xi, \eta) = \mathcal{F}[s_{m,n}(x, y)]. \tag{2}$$

Then, the complex amplitude $i_{m,n}(x, y)$ at imaging plane can be simply computed from $S_{m,n}(\xi, \eta)$ and system *CTF* (*sCTF*) according to Eq. (3), in which $\mathcal{F}^{-1}$ represents the inverse Fourier transform. *sCTF* determined by the optical system can be obtained with Eq. (4), in which d_i is the image distance, and D is the diameter of the lens. The amplitude of $i'_{m,n}(x, y)$ is updated from the captured raw image $i_m(x, y)$ with Eq. (5), and its spectrum $I'_{m,n}(\xi, \eta)$ is computed as shown in Eq. (6),

$$i_{m,n}(x, y) = \mathcal{F}^{-1}\left[S_{m,n}(\xi, \eta) \times sCTF\right], \tag{3}$$

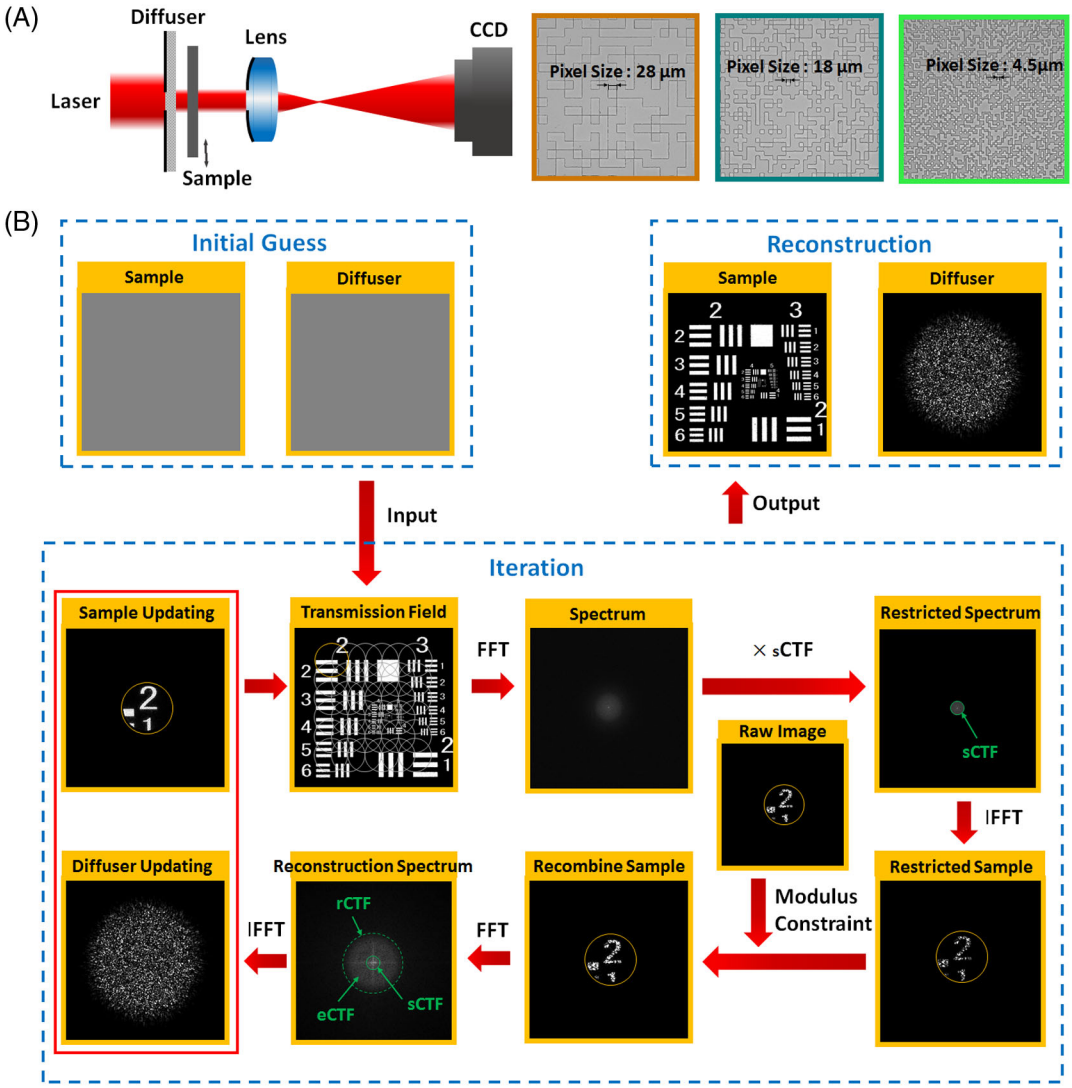

Fig. 2. (A) Scheme of speckle-illuminated Fourier ptychography and (B) flow chart of improved retrieval algorithm of speckle-illuminated Fourier ptychography. Random phase plates with different pixel sizes at 4.5, 18, and 28 μm, observed by bright-field microscope are listed in inset of (A).

$$sCTF(\xi,\eta) = \mathrm{circ}\left(\frac{\sqrt{\xi^2+\eta^2}}{D/2\lambda d_i}\right), \tag{4}$$

$$i'_{m,n}(x,y) = \sqrt{i_m(x,y)} \times \exp(\arg(i_{m,n}(x,y))), \tag{5}$$

$$I'_{m,n}(\xi,\eta) = \mathcal{F}[i'_{m,n}(x,y)]. \tag{6}$$

After updating $S_{m,n}(\xi,\eta)$ with Eq. (7), $S''_{m,n}(\xi,\eta)$ for the reconstructed object is the summation of $S'_{m,n}(\xi,\eta)$ in Eq. (7) within *sCTF* and $S_{m,n}(\xi,\eta)$ in Eq. (2) within expanded *CTF* (*eCTF*), according to Eq. (8). *sCTF* is a low-pass filter determined by the pupil of the optical system, and *eCTF* is an adjustable bandpass filter in the retrieval algorithm; thus, the reconstruction aperture (*rCTF*) is actually the summation of *sCTF* and *eCTF*, as shown in Eq. (9) and Figure 2(B). With a larger reconstruction aperture size, more higher-frequency components can be collected, and resolution enhanced $s''_{m,n}(x,y)$ can be computed with Eq. (10):

$$S'_{m,n}(\xi,\eta) = S_{m,n}(\xi,\eta) + \frac{|sCTF|}{|sCTF|_{\max}} \frac{sCTF^*}{[|sCTF|^2+1]} \times [I'_{m,n}(\xi,\eta) - S_{m,n}(\xi,\eta) \times sCTF], \tag{7}$$

$$S''_{m,n}(\xi,\eta) = S'_{m,n}(\xi,\eta) \times sCTF + S_{m,n}(\xi,\eta) \times eCTF, \tag{8}$$

$$rCTF = sCTF + eCTF, \tag{9}$$

$$s''_{m,n}(x,y) = \mathcal{F}^{-1}[S''_{m,n}(\xi,\eta)]. \tag{10}$$

Then, both the complex amplitudes of sample and probe light can be updated with Eqs. (11) and (12), respectively,

$$o'_{m,n}(x,y) = o_{m,n}(x,y) + \frac{|p_n(x,y)|}{|p_n(x,y)|_{\max}} \frac{p_n^*(x,y)}{[|p_n(x,y)|^2+1]} \times [s''_{m,n}(x,y) - p_n(x,y)\cdot o_{m,n}(x,y)], \tag{11}$$

$$p'_n(x,y) = p_n(x,y) + \frac{|o_{m,n}(x,y)|}{|o_{m,n}(x,y)|_{\max}} \frac{o^*_{m,n}(x,y)}{[|o_{m,n}(x,y)|^2 + 1]} \times [s''_{m,n}(x,y) - p_n(x,y) \cdot o_{m,n}(x,y)]. \tag{12}$$

After obtaining the updated sample and probe light in the mth sub-FoV, the above steps should be repeated again until all FoVs are addressed.

Finally, if the error shown in Eq. (13) is higher than 0.5%, another round of iteration will begin until the error is less than 0.5%,

$$\text{Error} = \frac{\sum \left| |o_{m,n+1}(x,y)| - |o_{m,n}(x,y)| \right|^2}{\sum |o_{m,n}(x,y)|}. \tag{13}$$

Different from the existing reconstruction algorithm of speckle-illuminated Fourier ptychography [31–35], the improved one, which considers the reconstruction aperture size, can further enhance the imaging quality (Fig. 1). Though larger reconstruction aperture size definitely can reconstruct more high-frequency components and accordingly generate images of higher resolution, it also introduces more noise and then decreases the imaging SNR. In the following sections, the influence of both diffuser roughness and reconstruction aperture size on imaging performance is studied numerically and experimentally.

3. NUMERICAL SIMULATIONS

Numerical simulations were first carried out to verify the improved retrieval algorithm for speckle-illuminated Fourier ptychography and then to analyze the influence of both diffuser roughness and reconstruction aperture size on imaging performance. Here, the illumination wavelength was set at 632.8 nm, a USAF 1951 resolution test chart was used as the sample, and random 0-π binary phase plates with different pixel sizes of 4.5, 18, and 28 μm were used to simulate diffusers of different roughnesses. According to Fig. 2(A), the introduced diffuser was located in front of the sample, and a lens with a focal length of 60 mm and an aperture diameter of 2 mm was set behind the sample; a CCD with the pixel size of 7.4 μm and the pixel number of 2048 × 2048 was used for raw image recording. After obtaining raw images, random noise was directly introduced to them to get the SNR of ∼ 20 dB. The distances of the sample to the diffuser and that to lens were 18 and 76 mm, respectively, and the distance between the lens and the CCD was 285 mm. All the parameters in numerical simulations were the same as those in experiments.

The influence of diffuser roughness on imaging performance was studied by using binary random phase plates with various pixel sizes and fixed reconstruction aperture (4.5 times of the *sCTF* in diameter). Figure 3(A) is a USAF 1951 resolution test chart, which was used as the object. Figure 3(B) shows the directly captured image without diffuser, in which the line pairs in Element 5 of Group 4 could be resolved, corresponding to a resolution of 39.40 μm. By using the random phase plate with a pixel size of 28 μm, 225 frames of raw images in different sub-FoVs were calculated; some of them are shown in Fig. 3(C). Since the diffuser is not very rough, there are fewer speckles in these raw images. Element 4 of Group 5 could be resolved in the reconstructed image, corresponding to a resolution of 22.00 μm, which is higher than that of a directly captured image. Moreover, by using a rougher random phase plate with a pixel size of 18 μm, there are more speckles in the computed raw images, as shown in Fig. 3(D). However, Element 6 of Group 5 could be resolved, revealing a resolution of 17.54 μm, which is higher than that of Figs. 3(B) and 3(C). While using a diffuser with the pixel size of 4.5 μm, an even higher resolution of 15.62 μm could be obtained, as Element 1 of Group 6 could be resolved in Fig. 3(E); however, the imaging SNR was significantly reduced. In order to quantify the imaging SNR, the peak SNR (PSNR) calculated from the red dotted field of interest (FoI) in Fig. 3(B) according to Eqs. (14) and (15) was adopted. In these equations, MSE is mean square error, $i(x,y)$ is the intensity of directly recorded image, which roughly has a uniform value of 1 within the area of the dotted square, $i'_n(x,y)$ is the reconstructed image after n iterations, $M \times N$ is image size, and MAX is the maximal intensity value of $i(x,y)$. The PSNRs of Figs. 3(B)–3(E) are 20.18, 17.83, 18.23, and 17.81 dB, revealing that a rougher diffuser can generate higher imaging resolution but more noise,

$$\text{PSNR} = 10 \times \log_{10} \frac{\text{MAX}^2}{\text{MSE}}, \tag{14}$$

$$\text{MSE} = \frac{1}{M \times N} \sum_x \sum_y [i(x,y) - i'_n(x,y)]^2. \tag{15}$$

Besides diffuser roughness, reconstruction aperture size in the retrieval algorithm also influences imaging performance. Using a fixed random phase plate with the pixel size of 18 μm and various reconstruction aperture diameters (2, 3, and 6 times the *sCTF* diameter), Element 6 of Group 4 in Fig. 3(F), Element 3 of Group 5 in Fig. 3(G), and Element 6 of Group 5 in Fig. 3(H) could be resolved, corresponding to resolutions of 35.00, 24.80, and 17.54 μm; this shows that with larger reconstruction apertures, higher imaging resolution could be obtained. The reconstructed images are listed in Figs. 3(F)–3(H), respectively. Additionally, the PSNRs of the reconstructed images in Fig. 3(F)–3(H) were computed as 19.10, 18.63, and 17.61 dB, revealing that larger reconstruction apertures also introduced more noise.

These numerical simulations verified that the improved retrieval algorithm considering the reconstruction aperture size can be well adopted for speckle-illuminated Fourier ptychography to provide superresolution images; in addition, it was also shown that a larger reconstruction aperture yields higher imaging resolution but lower imaging SNR, and a rougher diffuser results in higher imaging resolution but lower imaging SNR in general. In order to pursue the high-quality imaging performance of speckle-illuminated Fourier ptychography considering both imaging resolution and SNR, optimized diffuser roughness in the imaging system and reconstruction aperture size in the retrieval algorithm should be determined. Here, using a Siemens star as the sample for numerical simulations, Figs. 4(A) and 4(B) show the computed resolutions and

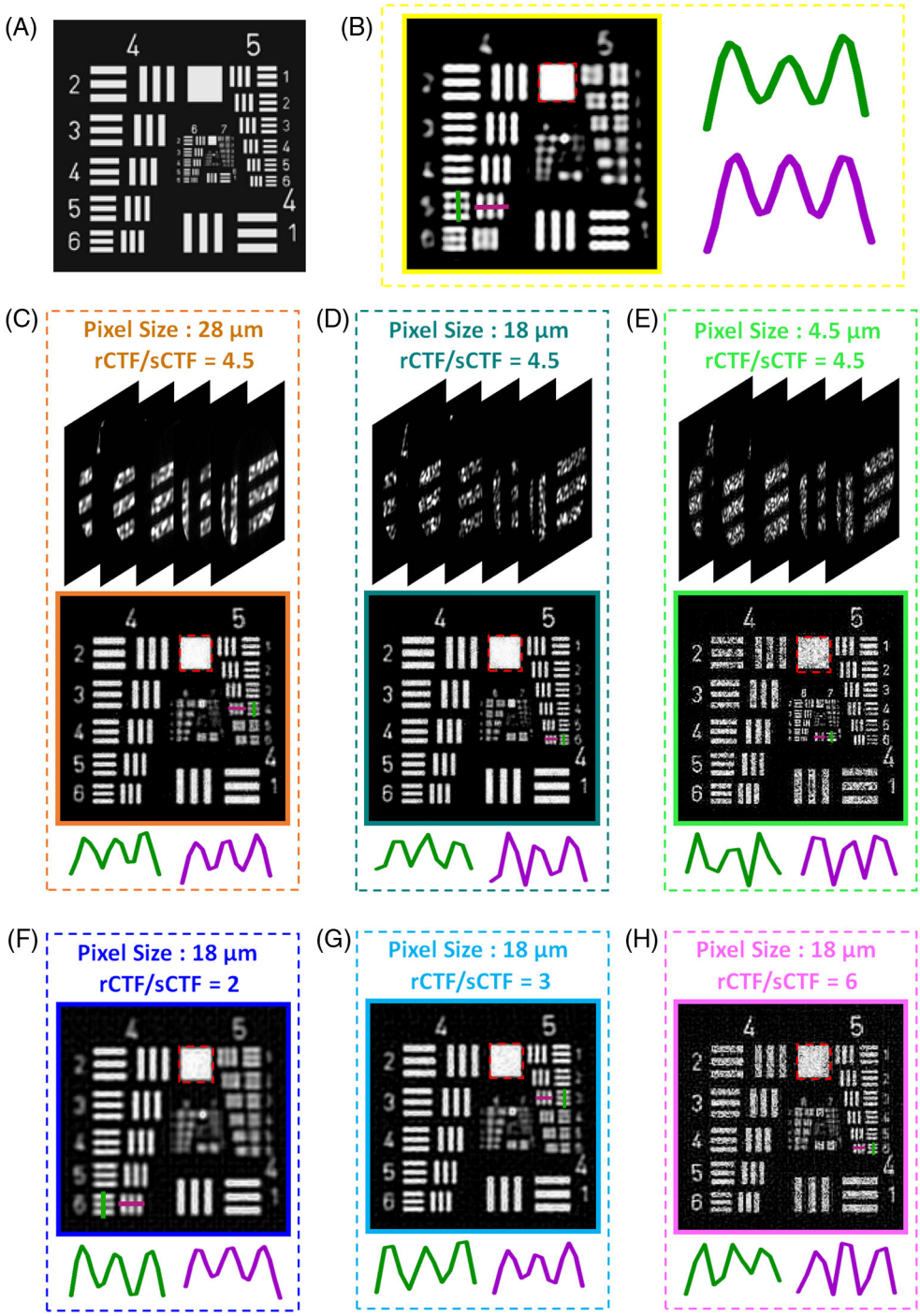

Fig. 3. Numerical simulations on speckle-illuminated Fourier ptychography. (A) Setting object; (B) directly captured image and resolved line pairs without diffuser; (C)–(H) raw images, reconstructed images, and resolved line pairs with different diffuser roughness and reconstruction aperture sizes.

PSNRs corresponding to various pixel sizes from 3 to 32 μm and different reconstruction aperture sizes from 1 to 6 times the *sCTF* diameter. With the diffuser pixel sizes and reconstruction aperture sizes adopted for simulations from Figs. 3(B)–3(G), six positions were marked in both Fig. 4(A) and 4(B) with tiny round spots of different colors. The resolutions, corresponding to these six round spots in Fig. 4(A), were 39.66, 21.42, 17.43, 15.64, 34.9, 25.18, and 17.15 μm, respectively, and the PSNRs corresponding to these six round spots in Fig. 4(B) were 20.09, 17.76, 18.17, 17.73, 19.08, 18.23, and 17.59 dB, respectively. All of these resolutions and PSNRs were quite close to that of the above simulations from Figs. 3(B)–3(G). Since more high-frequency components can be moved into the pupil of the optical system by using a rougher diffuser and can be properly reconstructed with a larger reconstruction aperture, we find in Fig. 4(A) that a larger reconstruction aperture and rougher

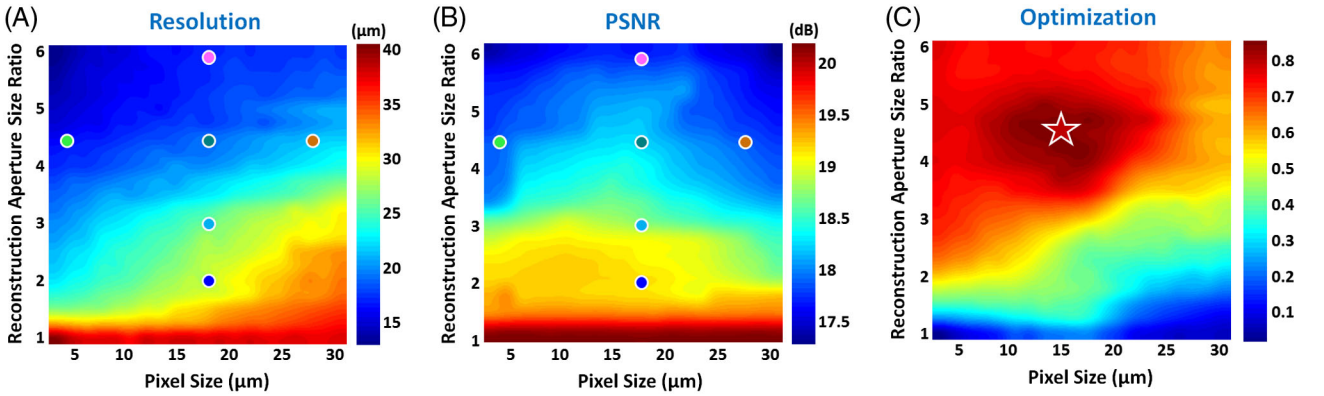

Fig. 4. Numerical simulations on speckle-illuminated Fourier ptychography in various diffuser roughness and reconstruction apertures. (A) Resolutions and (B) PSNRs with different pixel sizes and reconstruction aperture sizes; (C) optimization according to resolutions in (A) and PSNRs in (B); dots in (A) and (B) represent corresponding conditions in Fig. 3, and the star in (C) represents the optimized condition.

diffuser always result in higher resolution in general; however, the possibility of introducing wrong frequency components increases with larger reconstruction aperture; thus, in Fig. 4(B), we can find that higher PSNR is obtainable only for a rougher diffuser and smaller reconstruction aperture because the ratio of surely correct information inside the *sCTF* to that inside the *eCTF* is high for this case. For a larger aperture with a smoother diffuser, the correct information inside the *eCTF* is less, and the possibility of introducing wrong frequency components is high, resulting in a low PSNR. For a large aperture and too rough diffuser, it is very difficult to correctly reconstruct the diffuser with the proposed algorithm, and the inaccuracy in the reconstructed diffuser leads to low PSNR. Additionally, considering the trade-off between imaging resolution and PSNR, the optimized pixel size and reconstruction aperture size were simply determined. First, resolution and PSNR values in Figs. 4(A) and 4(B) were normalized as $\bar{d}$ and $\overline{\mathrm{PSNR}}$, respectively, in which small $\bar{d}$ means high resolution and large $\overline{\mathrm{PSNR}}$ represents high PSNR. Then, the ratio of $\frac{2}{3}(1-\bar{d})+\frac{1}{3}\overline{\mathrm{PSNR}}$ was computed, as shown in Fig. 4(C), and the weight of resolution is twice that of the PSNR. It is worth noting that both the evaluation function and the weights can be adjusted according to different requirements of both imaging resolution and PSNR. In other words, the optimized pixel size and reconstruction aperture size vary with different evaluation functions and weights. Finally, the optimized pixel size and reconstruction aperture size could be selected according to the maximal value in Fig. 4(C). Here, the optimized pixel size and reconstruction aperture size should be 15 μm and 4.7 times, respectively, to pursue high-quality imaging with imaging resolution of 17.23 μm and imaging PSNR of 18.20 dB. Moreover, according to Figs. 4(A) and 4(B), a larger reconstruction aperture results in higher resolution, but lower PSNR, while a rougher diffuser results in higher resolution, but not always lower PSNR, especially in the region with low roughness in Fig. 4(B). It is because PSNR was computed according to Eqs. (14) and (15), in which the reference is the original image. When the diffuser is rather smooth, high-frequency information cannot be collected; thus the error between the reconstructed image and the original image leads to reduced imaging PSNR.

4. EXPERIMENTS

In order to further verify the influences of both diffuser roughness and reconstruction aperture on the imaging performance of speckle-illuminated Fourier ptychography, experiments

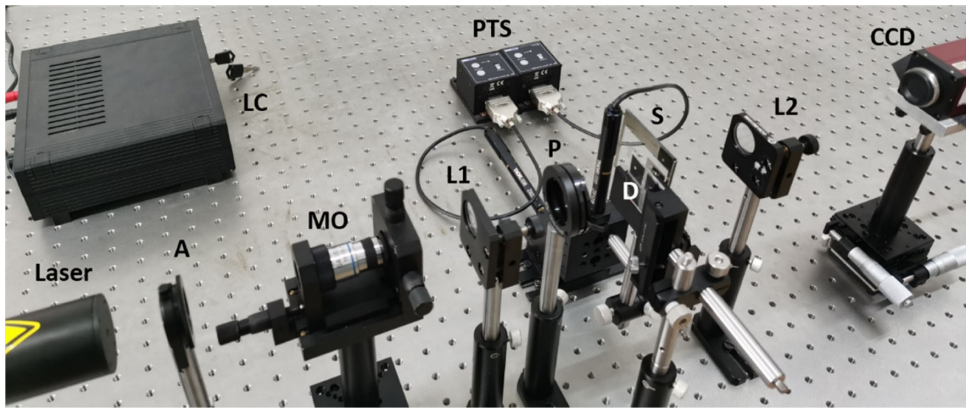

Fig. 5. Setup of speckle-illuminated Fourier ptychography. A, attenuator; MO, micro-objective; L1, lens1; P, pinhole; D, diffuser; S, sample; L2, lens2; LC, laser controller; PTS, precision translation stage.

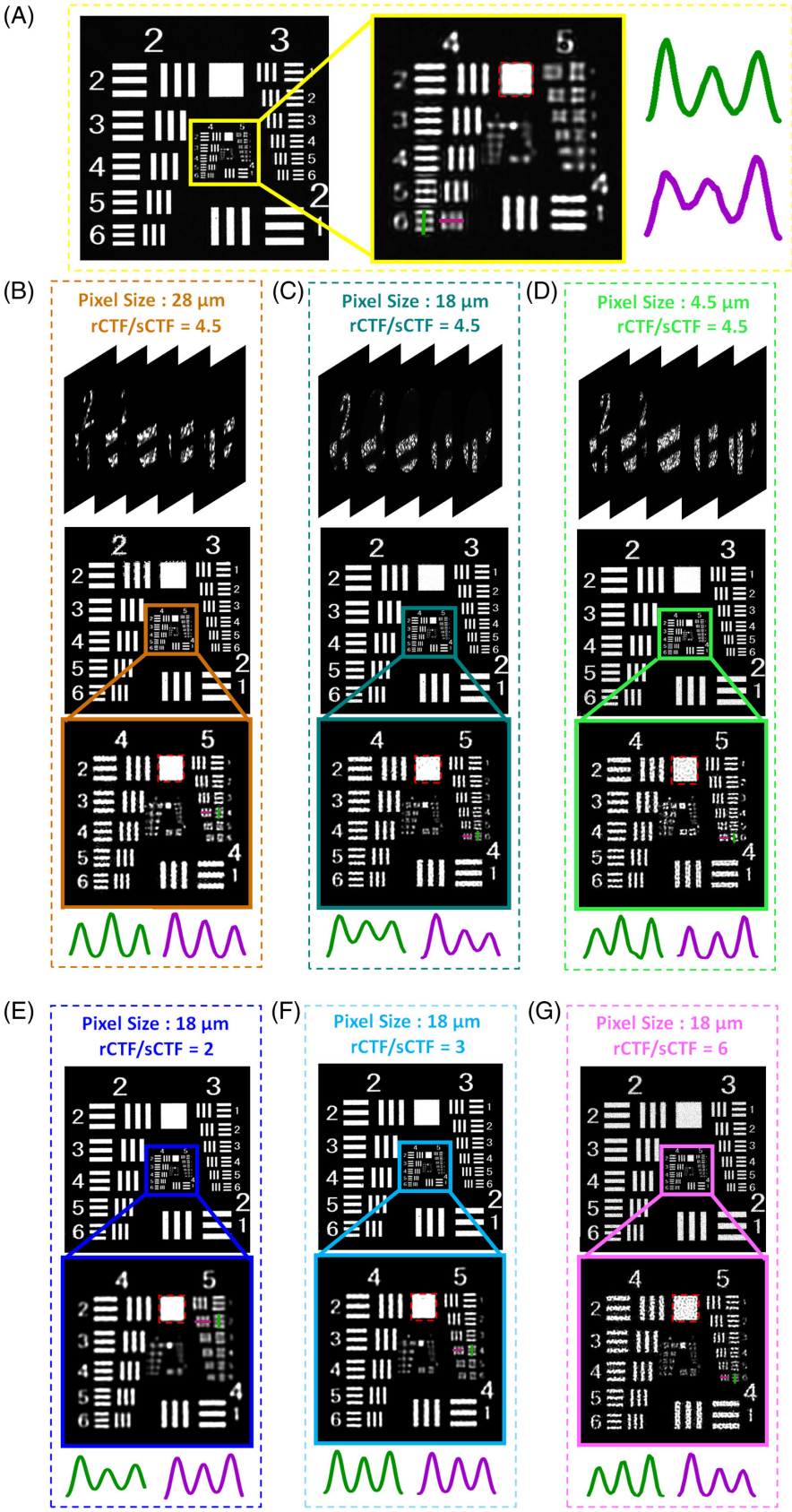

Fig. 6. Experiments of speckle-illuminated Fourier ptychography. (A) Directly captured image and resolved line pairs without diffuser; (B)–(G) raw images, reconstructed image, and resolved line pairs in conditions of different diffuser roughnesses and reconstruction aperture sizes.

Table 1. Average Contrast Values of Reconstructed Line Pairs in USAF 1951 Resolution Test Chart[a]

Figure	6(A)	6(B)	6(C)	6(D)	6(E)	6(F)	6(G)
Group 4, Element 5	0.85	0.99	0.99	0.99	0.98	0.99	0.97
Group 4, Element 6	***0.74***	0.99	0.99	0.99	0.89	0.98	0.96
Group 5, Element 1	0.36	0.98	0.99	0.99	0.75	0.97	0.94
Group 5, Element 2	0.25	0.94	0.98	0.96	***0.63***	0.85	0.91
Group 5, Element 3	0.21	0.82	0.97	0.91	0.28	0.79	0.88
Group 5, Element 4	0.13	***0.67***	0.94	0.85	0.15	***0.66***	0.85
Group 5, Element 5	0.09	0.30	0.86	0.79	0.10	0.26	0.82
Group 5, Element 6	0.03	0.23	***0.64***	***0.61***	0.04	0.18	***0.78***
Group 6, Element 1	0.02	0.19	0.25	0.10	0.02	0.13	0.24

[a]Line pairs are distinguishable when the contrast value is no less than 0.50.

were carried out by using the optical setup shown in Fig. 5. An expanded laser beam from an He–Ne laser (Sunpump, China) with the wavelength of 632.8 nm was used as the illumination. Binary 0-π random phase plates (fabricated by Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences) with different pixel sizes of 4.5, 18, and 28 μm were used as diffusers of different roughnesses. A USAF 1951 resolution test chart (Thorlabs, U.S.) was set behind the random phase plate with a separation of 18 mm. A lens (Daheng Optics, China) with a focal length of 60 mm and an aperture of 2 mm diameter was used to image the USAF 1951 on a CCD camera (Pike 421B, Allied Vision, Germany) with the pixel size of 7.4 μm and the pixel number of 2048 × 2048. The distance between the sample and imaging lens was 76 mm, and the distance between the imaging lens and CCD camera was 285 mm. The sample was fixed on a two-dimensional transition stage (Thorlabs, U.S.), and 225 raw images were captured by scanning 15 × 15 sub-FoVs covering the FoV of $\sim 4\,\text{mm} \times \sim 4\,\text{mm}$. The number of the collected raw images directly determines the size of the FoV and has no direct relationship to imaging resolution.

Figure 6(A) shows the directly captured image without a diffuser. Figures 6(B)–6(D) show the reconstructions with the same reconstruction aperture with diameter fixed as 4.5 times the *sCTF* diameter but with different random phase plates with pixel sizes of 4.5, 18, and 28 μm. Element 6 of Group 4 in Fig. 6(A), Element 4 of Group 5 in Fig. 6(B), Element 6 of Group 5 in Fig. 6(C) and Element 6 of Group 5 in Fig. 6(D) could be resolved corresponding to resolutions of 35.00, 22.00,

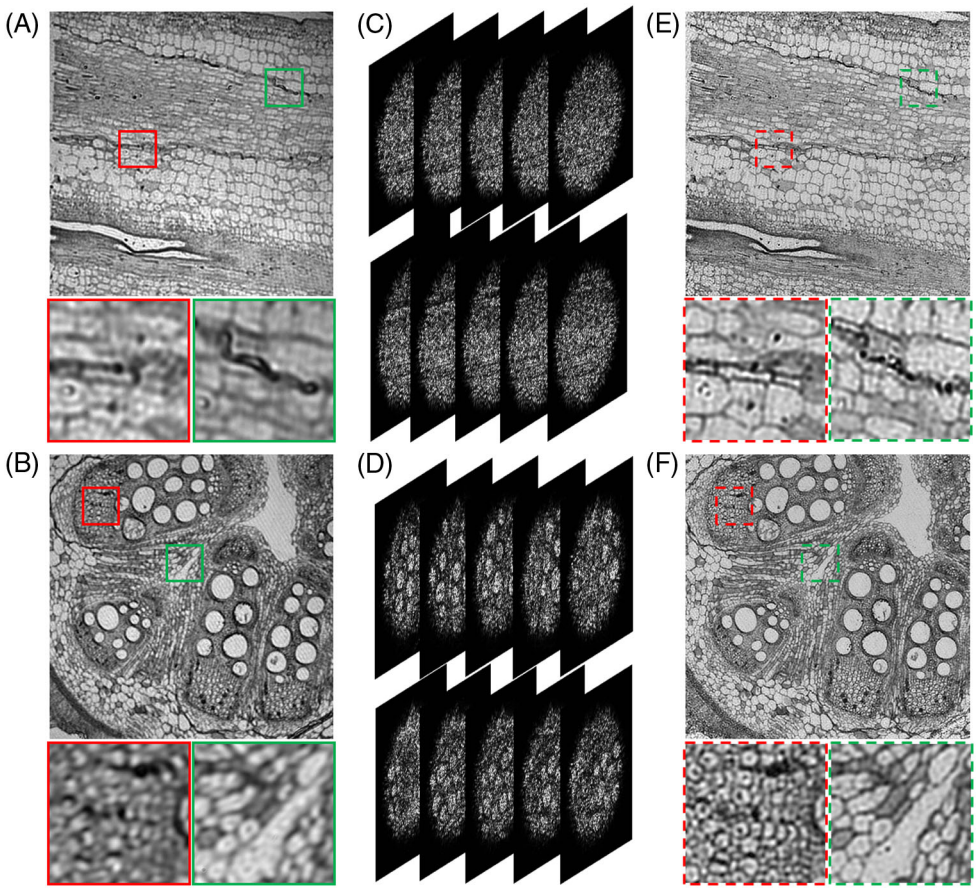

Fig. 7. Experiments of speckle-illuminated Fourier ptychography with optimized diffuser roughness and reconstruction aperture. (A)–(B) directly captured images; (C)–(D) raw images and (E)–(F) reconstruction images of pumpkin stem longitudinal and cross cuttings.

17.54, and 17.54 μm, respectively. The intensity of resolved elements was plotted as insets, and their computed contrasts are shown in Table 1, showing that with rougher diffusers, imaging resolution could be improved. However, PSNRs of these reconstructions calculated from red dotted FoIs were 20.24, 17.88, 18.32, and 17.86 dB, showing that more noise can be introduced by rougher diffusers in general. The low PSNR using a smooth diffuser with pixel size of 28 μm is mainly due to the loss of the high-frequency information in the reconstructed image. In addition, the influence of reconstruction aperture size on imaging performance was also experimentally analyzed. Figures 6(E)–6(G) reveal conditions using the same random phase plate with the pixel size of 18 μm but using different reconstruction aperture diameters 2, 3, and 6 times the *sCTF* diameter. Element 2 of Group 5 in Fig. 6(E), Element 4 of Group 5 in Fig. 6(F), and Element 6 of Group 5 in Fig. 6(G) could be resolved to correspond to resolutions of 27.80, 22.00, and 17.54 μm, according to both direct plots and computed contrast values in Table 1, revealing that with larger reconstruction apertures, imaging resolution can be improved. Additionally, also according to their PSNRs as 19.09, 18.72, and 17.65 dB estimated from the red dotted FoIs, more noise was also introduced when using larger reconstruction aperture sizes in the improved retrieval algorithm. Though differences existed in the reconstructed imaging in simulations and experiments, the experimental results, especially both imaging resolutions and PSNRs, fit well with the simulation results.

According to these experiments, it is proved that a rougher diffuser and larger reconstruction aperture yield higher imaging resolution but lower imaging SNR, coinciding with theoretical analysis and numerical simulations. Moreover, both imaging resolutions and PSNRs obtained in experiments are rather close to those computed in numerical simulations, indicating that the pixel size at 18 μm and reconstruction aperture size at 4.5 are close to the optimized setting and can be directly used in practical applications to pursue high-quality imaging considering both imaging resolution and SNR. According to these optimized parameters, pumpkin stem longitudinal and cross cuttings were measured, as shown in Fig. 7. Figures 7(A) and 7(B) are directly captured images without a diffuser. Using a diffuser with the pixel size of 18 μm (close to the optimized one) and the optimized reconstruction aperture diameter at 4.5 times the *sCTF* diameter, 225 raw images were captured by scanning the samples; part of them are shown in Figs. 7(C) and 7(D). Images as shown in Figs. 7(E) and 7(F) were reconstructed from Figs. 7(C) and 7(D), respectively. Compared to the FoIs in Figures 7(A)–7(B) and 7(E)–7(F), the speckle-illuminated Fourier ptychography could significantly improve resolution, since more details can be resolved in the reconstructed images in Figs. 7(E)–7(F) than in the directly captured images in Figs. 7(A)–7(B). More importantly, the experimental results prove that with comprehensive analysis on influence of both diffuser roughness in the imaging system and reconstruction aperture size in the retrieval algorithm on imaging resolution and SNR, optimized diffuser roughness and reconstruction aperture can be determined for high-quality imaging.

5. CONCLUSIONS

Besides the well-known effects of diffuser roughness on imaging performance in speckle-illuminated Fourier ptychography, we also find that reconstruction aperture size in the retrieval algorithm significantly influences the reconstructed image quality. In order to analyze speckle-illuminated Fourier ptychography comprehensively, to our best knowledge, it is the first time that the influence of both diffuser roughness in the imaging system and reconstruction aperture size in the retrieval algorithm on imaging performance, especially on imaging resolution and SNR, has been studied in detail. All the theoretical analysis, numerical simulations, and experiments are provided to prove that a rougher diffuser and a larger reconstruction aperture yield higher imaging resolution, but lower imaging SNR. Moreover, considering the trade-off between imaging resolution and SNR, both the diffuser roughness and the reconstruction aperture size can be optimized for high-quality imaging, proved by both simulations and experiments. Considering that this work provides a more comprehensive model of speckle-illuminated Fourier ptychography, we believe that this work can serve as useful guidance for high-quality imaging in speckle-illuminated Fourier ptychography.

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